

Ashutosh Shukla

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Bengaluru, India

Profile

Backend and Platform focused Software Engineer experienced in building self-hosted developer platforms, automating Linux deployments, and designing reliable distributed backend systems. I like working close to the metal — where deployment pipelines, system reliability, and developer experience actually matter.

Technical Skills

Languages: Python, TypeScript, JavaScript, Java, SQL

Backend: FastAPI, Node.js, Express, REST APIs, WebSockets, JWT, Microservices

Infrastructure & DevOps: Linux, Docker, Nginx, systemd, SSH, pyinfra, CI/CD, Git, GitHub Actions

Databases & Caching: PostgreSQL, Redis, MongoDB, Prisma, Connection Pooling, Query Optimization

Frontend: React.js, Tailwind CSS, HTML5

Experience

• Software Engineer Intern | LeapMile Robotics

January 2026 – Present | On-site

- Architected a shared PostgreSQL connection pool singleton and schema cache, eliminating per-request pool creation and reducing connection overhead across services.
- Re-engineered transaction lifecycle management and resolved production-critical database defects, improving reliability across robotic execution pipelines.
- Automated deployment of backend services to remote Linux servers using pyinfra, Nginx reverse proxy, and systemd service orchestration via SSH.

Projects

• Nexus – Self-Hosted PaaS

[\[GitHub\]](#)

- Built a production-grade self-hosted PaaS from scratch — automated the entire deploy pipeline from git push to live service, including stack detection, systemd orchestration, Nginx routing, and automatic rollback on failure.
- Developed real-time observability dashboard with system metrics, uptime checks, WebSocket-based journal log streaming, and Linux service control API.
- Validated pipeline across 4 stacks (FastAPI, Express, React and Next) achieving zero manual intervention from git push to production.

Stack: FastAPI, PostgreSQL, asyncpg, Nginx, systemd, React, WebSockets, Linux

• ChessMind – Real-time Chess Analysis Engine

[\[GitHub\]](#) [\[Live\]](#)

- Built real-time chess analysis platform with Stockfish, FastAPI, and WebSockets streaming asynchronous move evaluations via background threading.
- Implemented Redis caching with SHA-256 hashing, gracefully degrading under load and reducing repeated analysis latency by 80%.
- Designed fault-tolerant engine pool with **automated process recycling after 100 requests** preventing memory leaks in long-running sessions.

Stack: React, TypeScript, FastAPI, WebSockets, Python, Redis, Docker, Stockfish, Render, Vercel

• Aprameya – Full Stack Film Showcase Platform

[\[GitHub\]](#) [\[Live\]](#)

- Architected production-grade RESTful APIs in Node.js and Express with multi-layer caching reducing API response time by 25% under peak load.
- Built JWT-based authentication, role-based access control, and responsive React UI with advanced filtering and full-text search.

Stack: React, TypeScript, Node.js, Express, MongoDB, Tailwind CSS, JWT, Vercel

Achievements & Hackathons

- **Top 1%** – Tem-E-Thon 2025 Hackathon (3000+ participants); **Finalist** – VYUHA National Hackathon
- **2nd Place** – Sustainable Innovation Tech Competition, Skyline University (Dubai)
- **Innovation Patent (2025):** Smart Rainwater Harvesting Umbrella with Purification System (App. No. 202541065174)

Education

B.Tech in Computer Science Engineering

2022–2026

CMR Institute of Technology, Bengaluru | CGPA: 7.5/10 | Expected Graduation: June 2026

Coursework: Data Structures & Algorithms, Database Systems, Operating Systems, Machine Learning, Web Technologies

Interests & Languages

Backend Architecture, Distributed Systems, Applied AI, Competitive Programming, Chess

Languages: English, Hindi